

PRAHYAT PUANGJAKTHA

☎ 080-122-9289 ✉ pahyat.puangjaktha@gmail.com 💻 [pahyat puangjaktha](https://github.com/pahyat-pu) 🌐 [prayat-pu](https://prayat-pu.github.io)

Summary

Experienced Data Scientist with 4 years of specialized experience in data engineering, data science, deep learning, and machine learning. Proven expertise in natural language processing, predictive analytics, and computer vision, software engineering, underpinned by a solid academic background in computer science and statistics. Proficient in harnessing advanced tools such as Python, SQL, TensorFlow, PyTorch, and Spark to tackle complex business challenges and implement data-driven solutions. Skilled in working with satellite imagery and remote sensing data, applying machine learning techniques to extract insights and solve geospatial problems. Holds a Master's degree in Data Science from Mahidol University, Thailand. Known for a proactive approach to learning, problem-solving, and a passion for integrating the latest advancements in data science into practical applications.

Education

Mahidol University, Thailand

August 2021 – July 2023

Master of Science Program in Computer Science (Data Science)

CGPA: 3.63

Relevant Coursework: Advanced Artificial Intelligence, Data Science, Edge Computing & Internet of Things, Natural Language Processing, and Others

Chiang Mai University, Thailand

August 2016 – May 2020

Bachelor of Science (Computer Science)

CGPA: 3.54

Relevant Coursework: Pattern Recognition, Data Warehousing, Computer Vision, Artificial Intelligence, and Others

Technical Skills

Programming Language: Python, SQL, and Others

Data Science Framework: Conda, Docker, Numpy, Pandas, Scikit-Learn, Pytorch, knowledge graph, MLflow and others

Data Visualizing: Matplotlib, Seaborn, Pandas, Looker

Data Engineering: AWS (Glue, S3, Redshift, RDS, Athena), Azure (Data Factory, Synapse, Storage, Databricks), and Google Cloud Service (Storage, Big Query, Looker Studio, Cloud Composer), Apache Airflow, and Others

Experience

Synapse

August 2020 – October 2020

Machine Learning Engineer

Thailand

- Studied a variety of deep learning and machine learning models, and implemented them for specific tasks.
- Developed face and emotion detection models.
- Created web scraping scripts to download specific data for machine learning tasks.

National Astronomical Research Institute of Thailand (NARIT)

August 2019 – October 2020

Software Engineer Intern

Thailand

- Studied variable stars and developed a program to classify and characterize them using template fitting.
- The program calculates the period to convert Julian dates of data to phase, then fits the data with variable star templates, returning the fitting error score of all templates.

Geo-Infomatics and Space Technology Development Agency (GISTDA)

June 2019 – July 2019

Software Engineer Intern

Thailand

- Studied the APT and AX.25 protocols used in satellite communication between satellites and ground stations.
- Developed software for communication with a Raspberry Pi using Wi-Fi for the AX.25 protocol.
- Created software to transform signals from NOAA satellites into images for the APT protocol.

Projects

Analyzing A/B Testing | *Python, Pandas, Stat, and Sklearn* [GitHub](#)

- Conducted A/B testing for an e-commerce website to compare the conversion rates of the new and old pages.
- Applied statistical testing to determine the significance of differences between the old and new pages.

Handwritten Digits Classifier with PyTorch | *Python and PyTorch* [GitHub](#)

- Preprocessed a dataset for handwritten digit recognition using various normalization and augmentation techniques.
- Designed, built, and trained a neural network model using PyTorch to accurately recognize handwritten digits.

Landmark Classification & Tagging for Social Media | *Python, CNN, Transfer Learning, and PyTorch* [GitHub](#)

- Developed models to automatically predict the location of images based on landmarks depicted in them.

- Performed data preprocessing, designed and trained convolutional neural networks (CNNs), compared their accuracies, and deployed an application using the best-performing CNN.

Text Translation and Sentiment Analysis using Transformers | *Python, Transformers, HuggingFace, and PyTorch* [GitHub](#)

- Conducted sentiment analysis on movie reviews in English, French, and Spanish using transformer models.
- Translated French and Spanish movie reviews to English and applied sentiment analysis using the transformers in HuggingFace.

Face Generation | *Python, Generative Adversarial Network, and PyTorch* [GitHub](#)

- Implemented a generative adversarial network (GAN) to generate realistic images of faces.
- Developed a data pipeline for preprocessing, and created custom generator and discriminator models, along with a loss function and training loop for the GAN.

Predict Customer Churn | *Python, Sklearn, and Random Forest* [GitHub](#)

- Developed a machine learning pipeline to predict customer churn, encompassing data acquisition, exploratory data analysis, feature engineering, and model training and evaluation.
- Conducted a comparative analysis of Random Forest and Logistic Regression algorithms, implementing comprehensive tests for all functions within the pipeline.

ML Pipeline for Short-term Rental Prices in NYC

| *Python, Sklearn, Random Forest, Weight and Bias, and MLflow* [GitHub](#)

- Developed a machine learning pipeline to predict the average rental price of properties by analyzing prices of similar properties, ensuring the pipeline is reproducible and modular.
- Utilized tools such as Conda, Weights & Biases, MLflow, GitHub, and Python to facilitate EDA, data cleaning, data testing, model training, and the release and update of the pipeline.

A Dynamic Risk Assessment System | *Python, Flask, and Sklearn* [GitHub](#)

- Developed and implemented a machine learning model for risk assessment to predict the likelihood of clients leaving the company, enabling proactive client management.
- Implemented procedures for periodic model retraining, redeployment, monitoring, and evaluation to ensure reliable risk assessments and reduce client attrition.

Deploying an ML Model to Cloud Application Platform with FastAPI

| *Python, Random Forest, Sklearn, and FastAPI* [GitHub](#)

- Developed a classification model using publicly available Census Bureau data, created unit tests for model performance on various data slices, and deployed the model with FastAPI.
- Incorporated slice validation and API tests into a CI/CD framework using GitHub Actions to ensure continuous monitoring and improvement of the model.

Publication

“Paper, Meet Code”: A Deep Learning Approach to Linking Scholarly Articles With GitHub Repositories

| [paper](#)

2024

- Developed a method to automatically match academic articles with official software repositories.
- Utilized the SPECTER model to extract information from both academic articles and software repositories.
- Applied machine learning models, with the LightGBM model achieving the best performance in binary classification.

Evolving Compact Prediction Model for PM2.5 level of Chiang Mai | [paper](#)

2021

- Researched air pollution in Chiang Mai, focusing on PM2.5, a major contributor to health problems.
- Identified various factors influencing PM2.5 levels, highlighting fire as a significant correlating factor.
- Developed a machine learning model to predict PM2.5 levels in Chiang Mai for the next 3 hours.

Certifications

Certificate 1: [MLOps Fundamentals](#)

Certificate 2 : Complete Machine Learning and Data Science: Zero to Mastery

Certificate 3 : [Machine Learning DevOps Engineer](#)

Certificate 4 : [Data Engineering with AWS](#)

Certificate 5 : [Data Engineering with Microsoft Azure](#)

Certificate 6 : [Deep Learning](#)

Certificate 7 : [Statistics for Data Analysis](#)